Solver-Integrated Lossy and Lossless Compression for Scalable Flow Simulations

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THE STORAGE AND I/O WALL IN EXASCALE COMPUTING

Modern GPU-accelerated HPC simulations routinely generate datasets ranging from hundreds of gigabytes to **multiple terabytes** per snapshot, making storage, I/O, and data movement major bottlenecks in exascale scientific workflows. As compute capability continues to scale, filesystem bandwidth and storage infrastructure increasingly limit output cadence, visualization, and downstream analysis.

BOTTLENECKS

- **Multi-terabyte simulation snapshots**
- **Filesystem and MPI-I/O bottlenecks**
- **High storage and data transfer costs**
- **Limited output cadence at extreme scale**
- **Expensive post-hoc analysis workflows**

Result: I/O and storage wall limits science and productivity

OUR SOLUTION: SOLVER-INTEGRATED COMPRESSION

We integrate both **lossless** compression for restart/checkpoint reliability and **error-bounded lossy compression** for analysis-oriented outputs directly into the GPU-enabled spectral-element solver nekRS. By compressing solution fields in situ before MPI-I/O, we substantially reduce I/O volume and storage cost while preserving scientific value.

BENEFITS

- **Reduces storage footprint and I/O volume**
- **Minimizes data movement overhead**
- **Preserves QoIs within controlled tolerances**
- **Enables higher output cadence at extreme scale**
- **Maintains scientific usability of reconstructed fields**

Result: Lower I/O cost, higher output cadence, and scalable science

METHODOLOGY: COMPRESSION-AWARE I/O FRAMEWORK

A portable, compression-aware I/O framework integrated in CFD Solver

HPC Solver (nekRS)	• Spectral-element discretization • High-order accuracy • GPU-accelerated
Rank-local Fields (Element-based)	• Each rank holds a subset of elements • No communication for compression
Compression (in situ)	• Error-bounded SZ3 for analysis • Lossless Blosc2 for checkpoints
Portable Compressed Format (Self-describing)	• Rank-decomposable metadata • Decouples storage from MPI layout • Supports arbitrary read-time parallelism
Readers & Workflows (Coupled or Decoupled)	• Visualization · Post-hoc analysis • ROM · ML / GNN pipelines • In situ or ex situ access

QoI PRESERVATION

BOUNDS FROM POINTWISE TOLERANCE

Guaranteed pointwise error bound $|u_i - \tilde{u}_i| \leq \varepsilon \cdot \Delta u_{range}$

Closed form a priori bounds on physical QoI

Pick ϵ from target accuracy

GENERALIZATION: A PRIORI BOUNDS FOR ANY QoI

Quantity of Interest (QoI)	Error-bound scaling
Velocity magnitude $\lvert u\rvert$	$\leq \sqrt{3} \cdot \varepsilon \cdot \Delta u_{range}$
Kinetic energy $E_k = \frac{1}{2}\lvert u\rvert^2$	$O(\varepsilon \cdot u_{max}^2)$
Turbulence kinetic energy k	$O(\varepsilon \cdot u_{max}^2)$
Integral statistics $\langle\varphi\rangle$	$O(\varepsilon \cdot \lvert\varphi\rvert)$
Gradients ∇u, vorticity ω	$O(\varepsilon / h)$ → *grid-dependent*
Stress τ, Nusselt Nu	$O(\varepsilon / h)$

WORKED EXAMPLE: FUSION BLANKET VELOCITY

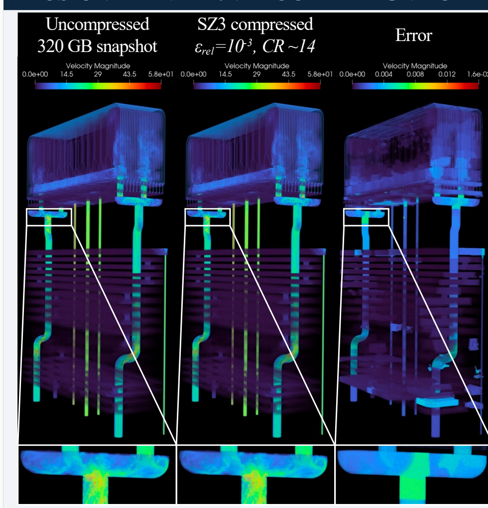
Given: $u_x, u_y, u_z \in [-58, 58]$, $\varepsilon = 10^{-3}$, $\Delta u_{range} = 116$

Per-component bound: $|u_i - \tilde{u}_i| \leq 0.116$

Magnitude bound: $|\,|u| - |\tilde{u}|\,| \leq \sqrt{3} \cdot \varepsilon \cdot \Delta u_{range} \approx$ **0.20** (worst case)

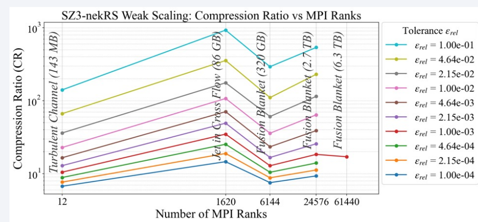
Observed: $|\,|u| - |\tilde{u}|\,| \leq 1.6 \times 10^{-2}$ → **~12× tighter than worst-case**

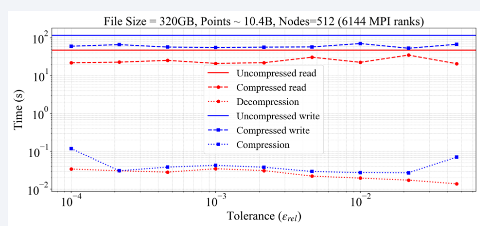
FUSION BLANKET: VELOCITY MAGNITUDE



Flow structures and features are preserved after compression.

PERFORMANCE: SCALING & I/O TIME





10-100× compression ratio and up to 8× I/O time reduction.

FUTURE WORK

- GPU-resident compression and asynchronous I/O
- SEM-aware transform-based compression strategies
- Integration with visualization tools (ParaView / VisIt)
- Evaluation of additional lossless and lossy compressors

Additional directions:

- Reactive and multiphysics flows
- Adaptive tolerance selection
- Compressed ML-ready datasets for ROM and AI workflows

GENERALIZATION

Applicable to a wide range of distributed scientific workflows with locally structured data layouts.

- High-order FEM / DG
- Structured-grid CFD
- Multiphysics workflows

Key ideas:

- Rank-local, in situ data reduction
- QoI-aware error-bounded compression
- Compressed parallel MPI-I/O

A general strategy for scalable, data-centric HPC workflows.

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Generative AI tools assisted with language editing and code/debugging tasks.

Data reduction is as critical as numerical computation in the exascale era.

REFERENCES & MORE

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